

Vector Function

A func. that takes in a parameter & returns a vector

e.g. Line eqn, $\mathbf{l} : \mathbf{r}(t) = \mathbf{r}_0 + t \mathbf{v}_1$, $t_1 < t < t_2$

Pos. vec. of point on $\mathbf{l}$ Direction vec. parallel to $\mathbf{l}$

Parameterisation aren't unique, the following is another line equation

$\mathbf{l} : \mathbf{r}(t) = \mathbf{r}_0 + \tan(s) \mathbf{v}_1$, $-\pi/2 < s < \pi/2$

This just scales $\mathbf{v}_1$

Tangent Vector

$$\frac{d\mathbf{r}}{dt} = \frac{d}{dt} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$= \frac{dx}{dt} \mathbf{i} + \frac{dy}{dt} \mathbf{j} + \frac{dz}{dt} \mathbf{k}$$

Unit tangent vector

$$\hat{\mathbf{t}} = \frac{\mathbf{r}'}{|\mathbf{r}'|} = \frac{\frac{d\mathbf{r}}{dt}}{\left| \frac{d\mathbf{r}}{dt} \right|}$$

Tangent Line at point P

$\mathbf{l}_{\text{tangent}} : \mathbf{w}(s) = \mathbf{P} + s \left. \frac{d\mathbf{r}}{dt} \right|_P$

Pos. vec. of P tangent vector at P

Normal vector

$$\mathbf{r}(u, v) = u \cos(v) \mathbf{i} + u \sin(v) \mathbf{j} + (4-u^2) \mathbf{k}$$

$$\mathbf{N} = \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v}$$

Unit Normal Vector

$$\hat{\mathbf{n}} = \frac{\mathbf{N}}{|\mathbf{N}|}$$

Parameterisation of common curves [Expected to know]

① Connecting Line b/w 2 pts

$$\mathbf{r}(t) = \mathbf{r}_1 + t(\mathbf{r}_2 - \mathbf{r}_1), \quad 0 \leq t \leq 1$$



$$\mathbf{r}(0) = \mathbf{r}_1$$

$$\mathbf{r}(1) = \mathbf{r}_2$$

single variable

② Circle

$$\mathbf{r}(t) = \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} = \cos t \mathbf{i} + \sin t \mathbf{j}, \quad 0 \leq t < 2\pi$$

width height

③ Ellipse If $a = b$, it's a circle

$$\mathbf{r}(t) = (a \cos t) \mathbf{i} + (b \sin t) \mathbf{j}, \quad 0 \leq t \leq 2\pi$$

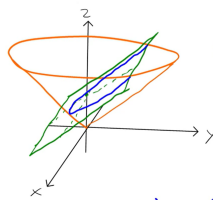
③ Any func. $y = f(x)$

$$\mathbf{r}(t) = \begin{pmatrix} t \\ f(t) \end{pmatrix} = t \mathbf{i} + f(t) \mathbf{j}$$

This just chooses x as the parameter, letting it vary thru y . Thus giving back the same graph

④ Intersection of 2 surfaces

$$\textcircled{4} \quad z = \sqrt{x^2 + y^2} \quad z = 1 + y$$



$$1 + y = \sqrt{x^2 + y^2}$$

$$1 + 2y + y^2 = x^2 + y^2$$

$$2y = x^2 - 1$$

$$y = \frac{x^2 - 1}{2}$$

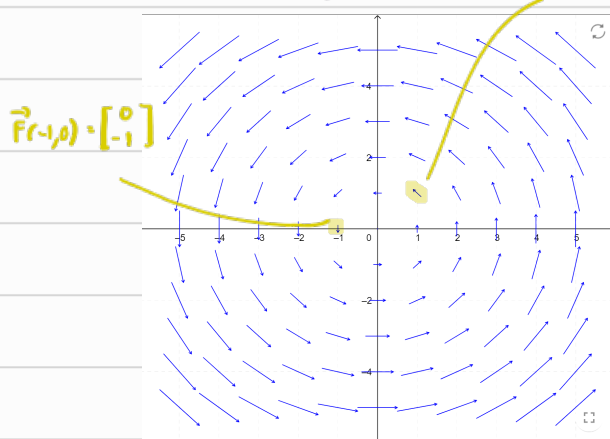
Let x be t

$$\mathbf{r}(t) = t \mathbf{i} + \frac{t^2 - 1}{2} \mathbf{j} + \frac{t^2 + 1}{2} \mathbf{k} \quad t \in \mathbb{R}$$

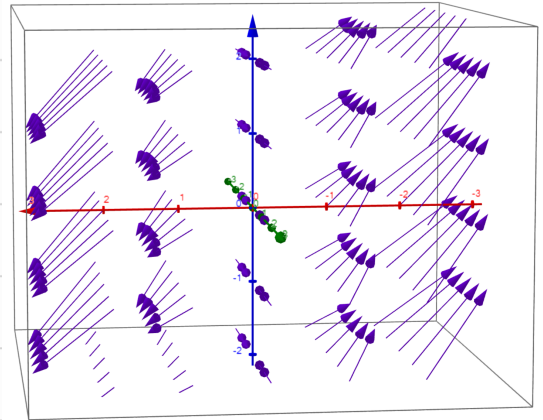
Vector Field

Function that assigns a vector to every point in a region

$$\vec{F}(x,y) = \begin{bmatrix} -y \\ x \end{bmatrix} \quad \vec{F}(1,1) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$



$$\vec{F}(x,y,z) = \begin{bmatrix} 2x \\ -2y \\ -2z \end{bmatrix} = 2x \hat{i} - 2y \hat{j} - 2z \hat{k}$$



Field Lines

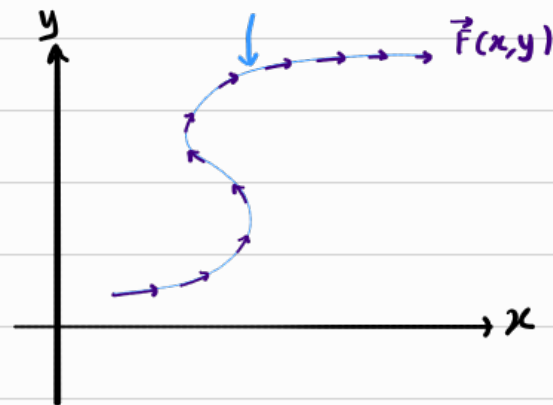
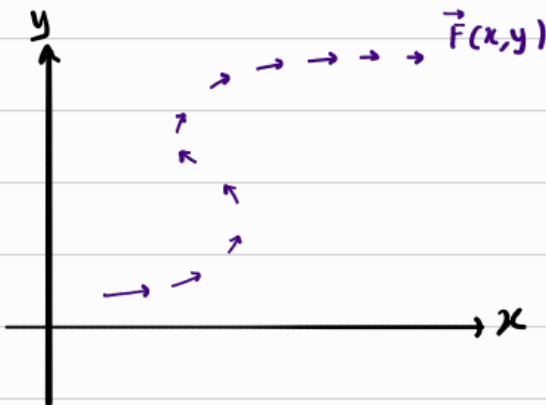
A field line is a smooth curve such that, at every point on the curve, the $\vec{r}'(t)$ tangent direc. agrees w the $\vec{F}(x,y)$ direc. of vector field there.

Notations

$\vec{r}(t) \Rightarrow$ position vector [A point]

$\vec{F}(x,y) \Rightarrow$ Takes a point, returns corresponding vector.

$$\text{Field Line, } \vec{r}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$



Assuming $\vec{r}(t)$ is a field line to vector field, $\vec{F}(x,y)$, then $\vec{r}'(t)$ is tangent to the field line & $\vec{F}(\vec{r}(t))$ is the field vector at the same point. Since a field line always follows the field direc.

$$\vec{r}'(t) = \vec{F}(\vec{r}(t))$$

Field Line Example

Given $\vec{F} = x^2 \hat{i} + y^2 \hat{j}$, find the eqn of the field line through $(-2, 2)$

Let field line, $\vec{r}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$

Since vector field is always tangent to field line,

$$\Rightarrow \vec{F}(\vec{r}(t)) = \vec{r}'(t)$$

$$\begin{bmatrix} x^2 \\ y^2 \end{bmatrix} = \begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix}$$

$$\therefore \frac{dx(t)}{dt} = x^2$$

$$\frac{1}{x^2} dx = dt$$

$$\int x^{-2} dx = \int 1 dt$$

$$-\frac{1}{x} = t + C_1$$

$$x(t) = \frac{-1}{t + C_1}$$

$$\frac{dy(t)}{dt} = y^2$$

$$\vdots$$

$$y(t) = \frac{-1}{t + C_2}$$

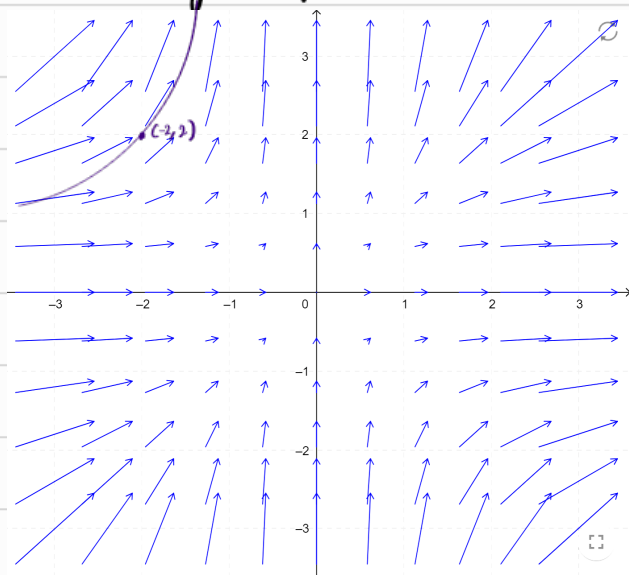
$$\therefore \vec{r}(t) = \begin{bmatrix} -1/(t + C_1) \\ -1/(t + C_2) \end{bmatrix}$$

Let $\vec{r}(0) = \begin{pmatrix} -2 \\ 2 \end{pmatrix}$ — we can do this cuz t is just a parameter

$$\begin{bmatrix} -1/C_1 \\ -1/C_2 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

$$\therefore C_1 = 1/2, C_2 = -1/2$$

$$\therefore \vec{r}(t) = \begin{bmatrix} -1/(t + 1/2) \\ -1/(t - 1/2) \end{bmatrix}$$



Field line as an eqn, $y = f(x)$

$$x = \frac{-1}{t + 1/2}$$

$$t = \frac{-1}{x} - \frac{1}{2}$$

$$y = \frac{-1}{t - 1/2}$$

$$= \frac{-1}{(-\frac{1}{x} - \frac{1}{2}) - \frac{1}{2}}$$

$$y = \frac{x}{1+x} \neq$$

$$\therefore y = \frac{x}{1+x}$$

Take note:

Since we are just looking for a field line eqn, $y = f(x)$, we only care about the overall shape, not how fast the parameter traces that shape.

So although $\vec{r}'(t) \neq \vec{F}(\vec{r}(t))$ but just $\vec{r}'(t) \parallel \vec{F}(\vec{r}(t))$ parallel, same direc., it's enough to give us the correct eqn. [e.g. $\vec{r}_1(t) = \begin{bmatrix} t \\ t \end{bmatrix}$ & $\vec{r}_2(t) = \begin{bmatrix} t^3 \\ t^3 \end{bmatrix}$ also trace the same line $y=x$]

Diff parametrization of the same curve may trace it w diff speed or orientation,

but after eliminating the parameter, they give the same geometric eqn of the curve

Alternative Method

Given $\vec{F} = x^2 \underline{i} + y^2 \underline{j}$, find the eqn of the field line through $(-2, 2)$

Let field line, $\vec{r}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$

Since vector field is always tangent to field line,

$$\Rightarrow \vec{F}(\vec{r}(t)) = \vec{r}'(t)$$

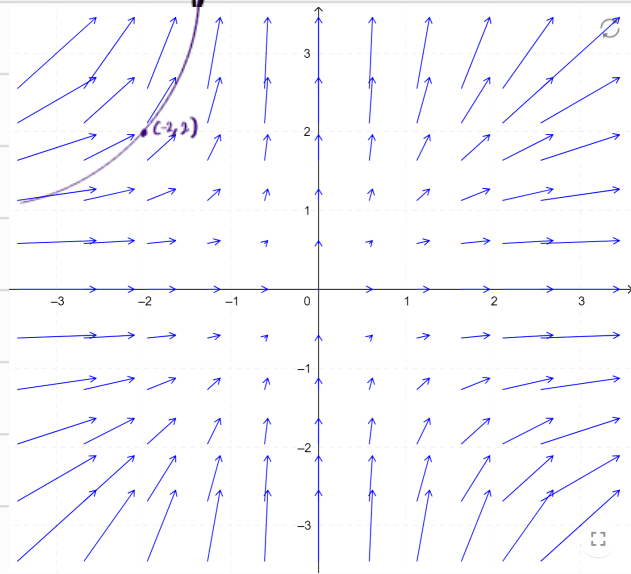
$$\begin{bmatrix} x^2 \\ y^2 \end{bmatrix} = \begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix}$$

$$\therefore \frac{dx(t)}{dt} = x^2 \text{ --- (1)}, \quad \frac{dy(t)}{dt} = y^2 \text{ --- (2)}$$

$$\frac{(2)/(1)}{(1)/(1)}, \quad \frac{dy}{dx} = y^2/x^2$$

$$\int \frac{1}{y^2} dy = \int \frac{1}{x^2} dx$$

$$-\frac{1}{y} = -\frac{1}{x} + C$$



At $(-2, 2)$,

$$-\frac{1}{2} = -\frac{1}{x} + C$$

$$C = -1$$

$$\therefore \frac{1}{y} = \frac{1}{x} + 1$$

$$y = \frac{x}{1+x} \quad \text{fx}$$

Take note:

Since we are just looking for a field line eqn, $y=f(x)$, we only care about the overall shape, not how fast the parameter traces that shape.

So although $\vec{r}'(t) \neq \vec{F}(\vec{r}(t))$ but just $\vec{r}'(t) \parallel \vec{F}(\vec{r}(t))$ parallel, same direc., it's enough to give us the correct eqn. [e.g. $\vec{r}_1(t) = \begin{bmatrix} t \\ t \end{bmatrix}$ & $\vec{r}_2(t) = \begin{bmatrix} t^3 \\ t^3 \end{bmatrix}$ also trace the same line $y=x$]

Diff parametrization of the same curve may trace it w diff speed or orientation,

but after eliminating the parameter, they give the same geometric eqn of the curve

Del Operator, ∇

∇ is a vector differential operator.

$$\nabla = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

Changing θ moves a point a circular arc, so the displacement in the θ -direc. is the arc length $r \cdot d\theta$. So to make sure its only $d\theta$, we $1/r$

① Gradient Vector ∇f Convert "per angle, $\partial f / \partial \theta$ " to "per distance"

$$\nabla f(x, y, z) = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

$$\nabla f(r, \theta, z) = \frac{\partial f}{\partial r} \hat{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{e}_\theta + \frac{\partial f}{\partial z} \hat{e}_z$$

$$\nabla f(r, \phi, \theta) = \frac{\partial f}{\partial r} \hat{e}_r + \frac{1}{r} \frac{\partial f}{\partial \phi} \hat{e}_\phi + \frac{1}{r \sin \phi} \frac{\partial f}{\partial \theta} \hat{e}_\theta$$

■ basis vectors

■ Normalising factors

• ∇f collects the gradient in the axis direc. for each coordinate

• $D_{\vec{u}} f = \nabla f \cdot \vec{u}$. $D_{\vec{u}} f$ is max when $\vec{u}$ goes in the direc. of ∇f

vector field / func.

② Divergence, $\nabla \cdot \mathbf{F}$

$$\mathbf{F}(x, y, z) = \begin{bmatrix} F_1(x, y, z) \\ F_2(x, y, z) \\ F_3(x, y, z) \end{bmatrix}$$

$$\nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

Sums up the local stretching rates in x, y, z direc.

e.g. $\partial F_1 / \partial x > 0$, this means right side pulls away faster, left side fills.

• Divergence measures the local tendency of the field at a point

to spread out from or converge toward that point.

[source]

$\therefore$ If $\nabla \cdot \mathbf{F} > 0 \Rightarrow$ Local expansion

[sink]

$\nabla \cdot \mathbf{F} < 0 \Rightarrow$ Local compression

$\nabla \cdot \mathbf{F} \neq 0 \Rightarrow$ Compressible

$\nabla \cdot \mathbf{F} = 0 \Rightarrow$ No volume change $\Rightarrow$ Incompressible

Other coordinate system

$$\mathbf{F}(r, \theta, z) = F_1 \hat{e}_r + F_2 \hat{e}_\theta + F_3 \hat{e}_z, \quad \hat{e}_r, \hat{e}_\theta, \hat{e}_z \text{ being basis vectors.}$$

$$\nabla \cdot \mathbf{F} = \frac{1}{r} \frac{\partial(r F_1)}{\partial r} + \frac{1}{r} \frac{\partial F_2}{\partial \theta} + \frac{\partial F_3}{\partial z}$$

③ Curl of vector field, $\nabla \times \mathbf{F}$

$$\mathbf{F}(x,y,z) = F_x \hat{i} + F_y \hat{j} + F_z \hat{k} = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \langle F_x, F_y, F_z \rangle$$

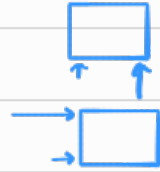
$$\nabla \times \mathbf{F} = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} \times \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \begin{bmatrix} \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \\ \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \\ \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \end{bmatrix}$$

Take z -component as an example: $\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}$

• If $\frac{\partial F_y}{\partial x} > 0$, upward "force" to the right is stronger, applying a counter CW torque

• If $\frac{\partial F_x}{\partial y} > 0$, right side "force" to the top is stronger, applying a CW torque

$\therefore$ If $\left| \frac{\partial F_y}{\partial x} \right| = \left| \frac{\partial F_x}{\partial y} \right|$, $\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} = 0$, no rotation will occur.



other coordinate system

Again, $\nabla \times \mathbf{F}$ is the same, regardless of coordinates

The curl of a vector field $\mathbf{F}(r, \theta, z) = F_r \mathbf{e}_r + F_\theta \mathbf{e}_\theta + F_z \mathbf{e}_z$ in cylindrical coordinates is

$$\nabla \times \mathbf{F} = \left(\frac{1}{r} \frac{\partial F_z}{\partial \theta} - \frac{\partial F_\theta}{\partial z} \right) \mathbf{e}_r + \left(\frac{\partial F_r}{\partial z} - \frac{\partial F_z}{\partial r} \right) \mathbf{e}_\theta + \left(\frac{1}{r} \frac{\partial}{\partial r} (r F_\theta) - \frac{1}{r} \frac{\partial F_r}{\partial \theta} \right) \mathbf{e}_z$$

The curl of a vector field $\mathbf{F}(r, \phi, \theta) = F_r \mathbf{e}_r + F_\theta \mathbf{e}_\theta + F_\phi \mathbf{e}_\phi$ in spherical coordinates is

$$\nabla \times \mathbf{F} = \left(\frac{1}{r \sin(\phi)} \frac{\partial}{\partial \phi} (\sin(\phi) F_\theta) - \frac{1}{r \sin(\phi)} \frac{\partial F_\phi}{\partial \theta} \right) \mathbf{e}_r + \left(\frac{1}{r \sin(\phi)} \frac{\partial F_r}{\partial \theta} - \frac{1}{r} \frac{\partial}{\partial r} (r F_\theta) \right) \mathbf{e}_\phi + \left(\frac{1}{r} \frac{\partial}{\partial r} (r F_\phi) - \frac{1}{r} \frac{\partial F_r}{\partial \phi} \right) \mathbf{e}_\theta$$

Laplacian Operator, ∇^2

[Multivariable version for 2nd derivative]

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\text{e.g. } \nabla^2 f = \nabla \cdot [\nabla f] = \nabla \cdot \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{bmatrix} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial f}{\partial z} \right)$$

The Laplacian measures the [divergence of the gradient] or [net local curvature] of a scalar field.

In other words, how the value at a point compares with the avg of nearby values.